TED UNIVERSITY

Department of Computer Engineering CMPE 491

SENIOR DESIGN PROJECT I
PROJECT SPECIFICATIONS REPORT

ReviewShield: A Multi-Signal Framework for Detecting Coordinated Fake Reviews in Turkish E-Commerce

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1. Introduction

This report presents the scope, objectives, and requirements of ReviewShield, a multi-signal framework designed to detect potentially coordinated fake reviews in Turkish e-commerce. It introduces the project's purpose and proposed approach, examines the constraints that may affect its development, and discusses the professional and ethical considerations involved. The report also reviews relevant research and existing detection methods, and defines the functional and non-functional requirements that will guide the system's design and implementation.

1.1 Description

Project title: ReviewShield: A Multi-Signal Framework for Detecting Coordinated Fake Reviews in Turkish E-Commerce

Product ratings and customer reviews influence purchasing decisions on e-commerce platforms. Review manipulation is often carried out by several accounts acting together: posting templated praise, targeting the same products, or producing synchronized bursts of reviews. Each of these reviews may look ordinary and fluent in isolation, so methods that examine one review at a time can miss coordinated activity. At the same time, legitimate events such as promotions or product launches can produce sudden increases in reviews that resemble manipulation. No single indicator is therefore conclusive.

ReviewShield is a platform-independent software prototype that identifies suspicious, potentially coordinated reviewing activity in Turkish e-commerce review data. Its primary unit of analysis is a group of reviewer accounts rather than a single review. The system combines three complementary signals: (i) semantic similarity of review texts, (ii) temporal anomalies such as review bursts and near-simultaneous activity, and (iii) the structure of a reviewer–product graph. Candidate groups are detected in the graph and ranked with an interpretable 0–100 suspicion score.

The intended workflow is batch-oriented. The user uploads a structured dataset (CSV or JSON); the system validates and preprocesses it, extracts text-similarity and temporal features, constructs a reviewer–product graph, detects candidate communities, computes group-level scores and presents the results in a simple web dashboard together with the supporting evidence. The suspicion score is a prioritization measure for human review; it is not proof of fraud and not a calibrated probability of dishonesty.

The project addresses the following research questions:

- Does combining text, temporal and graph signals improve the detection of coordinated groups compared with individual signals?
- How well does the approach recover controlled groups under different levels of textual similarity and temporal coordination?
- Which signals contribute to a flagged result, and which common legitimate behaviors generate false alarms?

The scope of the project, as defined in the Project Proposal, is summarized in Table 1.

Table 1. Project scope

Category	Items
In scope (minimum viable prototype)	Batch analysis of uploaded CSV/JSON datasets; data validation and conservative Turkish preprocessing; sentence-embedding based text similarity; rolling-window or statistical burst detection; reviewer–product graph and community detection; transparent weighted 0–100 group score with explanations; comparison of text-only, time-only, graph-only and combined configurations; a modest web dashboard with review summaries, suspicious groups and a graph view.
Stretch goals (only if data and time permit)	Learned fusion models and SHAP explanations (only with reliable labeled training data); near-duplicate indexing such as MinHash; PostgreSQL instead of SQLite if needed.
Out of scope	Real-time integration with e-commerce platforms; web scraping; graph neural networks; automated moderation or enforcement decisions; attributing intent to any person or account.

1.2 Constraints

This section discusses the constraints under which ReviewShield is designed, covering the economic, environmental, social, political, ethical, health and safety, manufacturability and sustainability aspects required by the course, followed by data and technical constraints that are specific to this project.

Economic constraints. The technologies planned in the Project Proposal (Python, pandas, NumPy, SciPy, Sentence Transformers / Hugging Face, NetworkX, scikit-learn, FastAPI, React and SQLite) are open-source. Text analysis is based on a suitable pretrained sentence-embedding model. Datasets must be publicly available or otherwise authorized for use. From the perspective of the application domain, review manipulation distorts purchasing decisions and competition between sellers; conversely, an incorrect flag could harm the reputation and revenue of a legitimate seller. For this reason the output is designed as a ranking for human inspection, not as an automatic decision.

Environmental constraints. ReviewShield is a software-only project; it involves no hardware manufacturing, materials or physical waste. Its environmental impact comes mainly from the electricity consumed by computation. The design limits this impact by using pretrained models only for inference, by processing data in batches rather than running a continuously active real-time service, and by restricting candidate edges in the account graph instead of comparing all account pairs.

Social constraints. Online reviews are a shared source of information for consumers, and coordinated manipulation erodes trust in e-commerce. A tool that helps analysts find suspicious groups can support consumers and honest sellers. However, the same analysis can produce false positives: customer communities, promotional campaigns, product launches or viral events may create activity that resembles coordination. Wrongly labeling real customers or sellers as fraudulent would be socially harmful. The system therefore presents evidence, keeps a human in the loop, avoids public accusations and includes an explicit analysis of false alarms.

Political and legal constraints. The project operates in a regulated area. In Turkey, consumer reviews are governed by the Regulation on Commercial Advertising and Unfair Commercial Practices, issued under Law No.

6502 on Consumer Protection. An amendment to this regulation was published in the Official Gazette on 1 July 2026 (No. 33297) and entered into force on 1 August 2026; it revises the rules on the verification, publication and ordering of online consumer reviews (Erdem & Erdem, 2026). The Advertising Board has since imposed access bans on services that sell fake ratings and reviews (Para Dergi, 2026). In the European Union, Directive (EU) 2019/2161 added the submission and commissioning of false consumer reviews to the list of unfair commercial practices. Personal data processing in Turkey is subject to Law No. 6698 on the Protection of Personal Data (KVKK). These frameworks are taken into account in the project's data handling; the project remains neutral towards platforms, sellers and brands, and its outputs are not presented as legal findings.

Ethical constraints. The main ethical constraints are the risk of mislabeling innocent users, the protection of privacy, the lawful use of data and honest reporting of results. These are discussed in detail in Section 1.3.

Health and safety constraints. The project involves no physical equipment, hazardous materials or health data, and therefore poses no direct physical health or safety risk. The relevant risk is indirect: reputational or psychological harm to individuals or sellers who could be wrongly suspected of manipulation. This risk is mitigated by using anonymized account identifiers, by not publishing identifiable results, and by stating that the score is not proof of fraud.

Manufacturability constraints. For a software project, manufacturability is interpreted as the ability to build, reproduce and deploy the system reliably. ReviewShield relies on widely used, well-documented libraries, a fixed input schema, a modular pipeline and recorded configurations (thresholds, time windows, score weights and model versions) so that the system can be rebuilt and its results reproduced on another machine. The batch-oriented design keeps the prototype manageable for a four-person team.

Sustainability constraints. The system must remain maintainable and extendable after the course. Components such as the embedding model or the community-detection algorithm are designed to be replaceable without redesigning the whole pipeline. The platform-independent input schema allows the system to be reused with other review datasets. Source code, documentation, the evaluation dataset protocol and reproducible notebooks are planned deliverables, so that the work can be continued, verified or extended.

Data constraints. Complete Turkish review datasets that include account identifiers, product identifiers and timestamps may be scarce. Existing Turkish resources such as the deception detection dataset of Akkol and Gökşen (2024) were designed for text-level analysis, and their suitability for group-level analysis must be checked. An early dataset audit will examine field completeness and licensing. If Turkish ground-truth labels are insufficient, controlled synthetic coordinated groups will be inserted into a background of real, legally obtained Turkish reviews, and results on real and synthetic data will be reported separately. Missing fields will be documented rather than inferred.

Technical constraints. Dense account–account projections can become computationally expensive on large datasets; candidate edges must therefore be restricted using product overlap and thresholds. Semantic similarity does not by itself imply deception, and review bursts can be legitimate; the system must combine independent evidence rather than rely on a single indicator. Text normalization must be conservative so that negation, meaningful spelling and sentiment cues are not removed.

1.3 Professional and Ethical Issues

The team's professional and ethical responsibilities are based on the ACM Code of Ethics and Professional Conduct (ACM, 2018), the Software Engineering Code of Ethics and Professional Practice (ACM/IEEE-CS Joint Task Force, 1999) and the IEEE Code of Ethics (IEEE, n.d.). Because ReviewShield analyzes the behavior of people and may draw attention to accounts as suspicious, these responsibilities directly affect its design. Table 2 summarizes the main issues and how the project addresses them; the paragraphs below explain them in more detail.

Table 2. Professional and ethical issues and how they are addressed

Issue	Relevant principle	How ReviewShield addresses it
Harm from false accusations	ACM 1.2 Avoid harm; SE Code Principle 1 (Public)	Score is a prioritization measure, not a verdict; no automated enforcement; evidence is shown for human review; false alarms are analyzed.
Honest reporting of results	ACM 1.3 Be honest and trustworthy; IEEE: honest and realistic claims based on available data	Real and synthetic results reported separately; no predetermined performance claims; limitations and errors reported; no claim of being the first such method.
Fairness	ACM 1.4 Be fair and take action not to discriminate	Only behavioral and textual review data are used, no demographic attributes; legitimate behaviors that trigger false alarms are examined.
Privacy and personal data	ACM 1.6 Respect privacy; KVKK (Law No. 6698)	No personally identifying data required; anonymized account identifiers; no identifiable results published.
Lawful data use and credit	ACM 1.5 Respect the work required to produce new ideas…; ACM 2.3 Know and respect existing rules; ACM 2.8 Access resources only when authorized	Only authorized, licensed datasets; no live scraping; dataset provenance, licenses, models and prior work are cited.
Quality and competence	ACM 2.1 Strive to achieve high quality; ACM 2.5 Give comprehensive evaluations including risks; SE Code Principle 3 (Product)	Validation of the embedding model on Turkish examples, sensitivity testing of thresholds, baseline and ablation comparisons, reproducible experiments.
Responsibilities within the team	SE Code Principles 5 (Management), 7 (Colleagues) and 8 (Self)	Tasks and responsibilities recorded in the backlog; shared responsibility for integration and documentation; proper credit for contributions.

Avoiding harm and presumption of innocence. The most serious risk of the project is that a legitimate reviewer or seller is treated as fraudulent. ReviewShield therefore flags patterns, not intentions. Its output ranks groups for inspection, displays the underlying evidence and explicitly states that the score is not proof of fraud. The system does not make moderation or enforcement decisions, and the team will not make public accusations about any real account, seller or platform.

Honesty in claims and results. The team will report results honestly and realistically. Results obtained on synthetic injected groups will not be presented as real-world fraud detection accuracy. Labels in public datasets, such as those produced by a platform's own review filter, are proxies and their limitations will be stated. Performance will not be promised in advance; the evaluation emphasizes a reproducible protocol and candid error analysis. The contribution is described as an integration that will be tested experimentally, not as the first or best method.

Fairness. Coordination signals can also arise from legitimate behavior such as promotions, product launches or customer communities. The team will examine false-positive examples to understand which legitimate behaviors trigger alarms, and will not use demographic or other sensitive attributes in scoring.

Privacy and data protection. The system is designed to work with anonymized account identifiers and does not require names, contact details or other personally identifying data. Reports, the webpage and presentations will contain only aggregated or anonymized examples. Data handling will be reviewed in accordance with applicable data protection requirements, including KVKK, before datasets are used.

Lawful use of data and intellectual property. Only datasets that are publicly available for research or otherwise authorized will be used, and their licenses and terms will be respected. Live scraping of e-commerce platforms is excluded. Datasets, pretrained models, libraries and prior research will be properly cited.

Professional competence and quality. The team will validate its methods before relying on them, for example by checking the selected embedding model on manually inspected Turkish examples before choosing similarity

thresholds and by sensitivity-testing time windows. Single-signal baselines and ablations ensure that the contribution of each signal is measured rather than assumed. Configurations will be recorded so that results can be reproduced and reviewed by the supervisor and jury.

Teamwork and academic integrity. Responsibilities are shared among the four team members, and the allocation of tasks is recorded in the project backlog. Each member is responsible for the quality of their own contributions, for supporting the others and for giving proper credit. All reports will be the team's own work, with all sources cited.

2. Literature Review

This section expands the preliminary literature review of the Project Proposal. Prior work is grouped by the type of evidence it uses: review text, temporal and behavioral patterns, group-level patterns and graph structure. Enabling techniques and candidate datasets are then summarized, and the section ends with the gap that ReviewShield addresses.

2.1 Text-based detection

Jindal and Liu (2008) conducted one of the first large-scale studies of opinion spam using Amazon reviews and identified duplicate and near-duplicate reviews as a strong indicator of spam. Ott et al. (2011) built a gold-standard dataset of deceptive hotel reviews written by crowd workers and truthful reviews, and showed that classifiers using n-gram and psycholinguistic features could detect the deceptive reviews with accuracy close to 90%. However, Mukherjee et al. (2013) found that the same linguistic features were much less effective on real reviews filtered by Yelp, while behavioral features of reviewers were considerably more informative. For Turkish, Akkol and Gökşen (2024) created a deception detection dataset of 5,013 hotel reviews containing real reviews, human-written fake reviews and AI-generated fake reviews. These results indicate that text is useful, especially for detecting near-duplicate or templated content, but that text alone is insufficient on real data. In ReviewShield, text is therefore used as one signal (semantic similarity between reviews) rather than as a stand-alone classifier.

2.2 Temporal and behavioral signals

Xie et al. (2012) showed that singleton review spam, which is hard to detect from individual reviews, can be detected through abnormal bursts in the time series of a store's reviews and ratings. Fei et al. (2013) observed that spammers tend to appear in review bursts, modeled reviewers within bursts as a network and inferred spammers collectively. These studies support the use of burst detection, but they also note that bursts can result from legitimate causes such as sudden popularity. ReviewShield therefore treats bursts and near-simultaneous activity as evidence that must be combined with other signals, not as proof.

2.3 Group-level detection

Mukherjee et al. (2012) focused on fake reviewer groups. They used frequent itemset mining on reviewer–product data to generate candidate groups, defined group and individual spam indicators such as posting within a short time window, rating deviation and content similarity, and ranked groups with a relation-based model (GSRank). Their work shows that groups can be more damaging and more detectable than individual spammers. ReviewShield adopts the group as the unit of analysis, but generates candidate groups through a weighted account graph and community detection, and measures content similarity with multilingual sentence embeddings suitable for Turkish.

2.4 Graph-based detection

Wang et al. (2011) modeled reviewers, reviews and stores as a heterogeneous review graph and computed trustiness, honesty and reliability scores iteratively. Akoglu et al. (2013) proposed FraudEagle, an unsupervised method that models the reviewer–product network as a signed bipartite graph and uses loopy belief propagation. Rayana and Akoglu (2015) extended this idea with SpEagle, which combines metadata (text, behavior and

timestamps) with the review network, and introduced the YelpNYC and YelpZip datasets. Hooi et al. (2016) proposed FRAUDAR, a dense-subgraph detection method designed to resist camouflage by fraudulent accounts. More recent work uses graph neural networks; for example, CARE-GNN (Dou et al., 2020) improves robustness against camouflaged fraudsters on Yelp and Amazon review graphs. Yu et al. (2022) survey graph learning methods for fake review detection. These studies show that relational structure is a strong signal. Graph neural networks, however, typically require reliable labeled data, which is scarce for Turkish; they are therefore outside the scope of the ReviewShield prototype.

2.5 Enabling techniques

Sentence-BERT (Reimers & Gurevych, 2019) produces sentence embeddings that can be compared efficiently with cosine similarity, and multilingual knowledge distillation (Reimers & Gurevych, 2020) extends such models to many languages, including Turkish. These methods enable the semantic comparison of reviews that are paraphrased rather than copied. The Louvain method (Blondel et al., 2008) detects communities by optimizing modularity and scales to large networks; it is the initial choice for detecting candidate groups in the account graph, with other feasible algorithms as alternatives.

2.6 Candidate datasets

Table 3 lists datasets identified so far. Their availability, licenses and field completeness will be confirmed in the dataset audit before use.

Table 3. Candidate datasets

Dataset	Language / domain	Relevant content	Use in ReviewShield
Deceptive opinion spam corpus (Ott et al., 2011)	English / hotel reviews	Review text with crowdsourced deceptive and truthful labels	Text-signal checks only; no reviewer–product–time structure
YelpChi (Mukherjee et al., 2013)	English / restaurant and hotel reviews	Reviewer, product, rating, date, text; labels from Yelp's filter	Multi-signal generalization check; labels are platform proxies
YelpNYC and YelpZip (Rayana & Akoglu, 2015)	English / restaurant reviews	Reviewer, product, rating, date, text; labels from Yelp's filter	Multi-signal generalization check; labels are platform proxies
Turkish deception detection dataset (Akkol & Gökşen, 2024)	Turkish / hotel reviews	Real, human-written fake and AI-generated fake review texts	Turkish text-similarity validation; suitability for group analysis to be checked

2.7 Summary and research gap

Table 4. Comparison of related approaches with ReviewShield

Work	Main signals	Unit of analysis	Relation to ReviewShield
Jindal & Liu (2008); Ott et al. (2011)	Text	Review	Text alone is insufficient on real data; used as one signal
Xie et al. (2012); Fei et al. (2013)	Time / bursts, behavior	Review, reviewer	Temporal coordination signal
Mukherjee et al. (2012)	Behavior, time, content similarity	Group	Group as unit of analysis
Wang et al. (2011); Akoglu et al. (2013); Rayana & Akoglu (2015); Hooi et al. (2016)	Graph, metadata	Reviewer, review, product	Reviewer–product graph and community structure
Dou et al. (2020); Yu et al. (2022)	Graph learning (GNN)	Node	Requires labels; out of scope

Work	Main signals	Unit of analysis	Relation to ReviewShield
ReviewShield	Semantic text similarity, temporal anomalies and graph communities combined	Group	Explainable 0–100 group score for Turkish e-commerce reviews

Prior work shows that text, temporal and graph evidence are each informative, that group-level analysis is valuable, and that text-only methods weaken on real data. Most of these studies, however, use English datasets, and Turkish resources are mainly text-oriented. ReviewShield therefore tests whether an explainable, group-oriented combination of multilingual semantic similarity, temporal coordination and graph communities is useful for Turkish e-commerce reviews. The project does not claim to be the first such method or to guarantee superior performance; this combination will be evaluated experimentally against single-signal baselines.

3. Requirements

This section identifies the initial requirements of ReviewShield. Priorities follow the MoSCoW scheme: Must (part of the minimum viable prototype), Should (planned, but the prototype remains usable without it), and Could (stretch goal). Conditional is used separately for a requirement that becomes mandatory only if its stated condition occurs. Each requirement has a planned verification method: T (test), D (demonstration), I (inspection) or A (analysis or experiment).

3.1 Users

The primary user of the system is an analyst who uploads a review dataset, runs the analysis and inspects the flagged groups and their evidence.

3.2 Functional requirements

Table 5. Data import, validation and preprocessing

ID	Requirement	Priority	Verif.
FR-01	The system shall allow the user to upload a review dataset in CSV or JSON format.	Must	T
FR-02	The system shall validate the records against the input schema (Section 3.4) and report missing, empty or malformed fields.	Must	T
FR-03	The system shall accept optional additional metadata fields without failing.	Should	T
FR-04	If a field needed by an analysis (e.g., timestamp or account ID) is missing, the system shall skip or limit that analysis, report the limitation, and shall not infer the missing values.	Must	T
FR-05	The system shall display a summary of the imported dataset (e.g., number of reviews, accounts and products, time span and field completeness).	Should	D
FR-06	The system shall apply conservative Turkish text normalization that preserves negation, meaningful spelling and sentiment cues.	Must	I

Table 6. Text-similarity analysis

ID	Requirement	Priority	Verif.
FR-07	The system shall convert each review text into a vector using a Turkish-compatible or multilingual pretrained sentence-embedding model.	Must	T
FR-08	The system shall compute cosine similarity between reviews of the same product and between reviews written by accounts in a candidate group.	Must	T
FR-09	Similarity thresholds shall be configurable and shall be chosen after validating the model on a manually inspected sample of Turkish reviews.	Must	A

ID	Requirement	Priority	Verif.
FR-10	The system could use near-duplicate indexing (e.g., MinHash) to speed up similarity search.	Could	T

Table 7. Temporal analysis

ID	Requirement	Priority	Verif.
FR-11	The system shall aggregate review activity per product over configurable time windows.	Must	T
FR-12	The system shall detect review bursts using rolling-window counts or a statistical deviation from a local baseline, with documented thresholds.	Must	T
FR-13	The system shall compute a separate coordination signal for near-simultaneous review activity of multiple accounts.	Must	T
FR-14	The system shall support sensitivity testing of time windows and thresholds.	Should	A

Table 8. Graph analysis and candidate groups

ID	Requirement	Priority	Verif.
FR-15	When account and product identifiers are available, the system shall construct a bipartite graph connecting anonymized accounts to the products they reviewed.	Must	T
FR-16	The system shall construct a weighted account–account graph from co-reviewed products, with edge weights incorporating available review timing and text similarity. If either signal is unavailable, the system shall document the limitation rather than invent missing evidence.	Must	T
FR-17	The system shall reduce the influence of extremely popular products and restrict candidate edges using product overlap and thresholds.	Must	T
FR-18	The system shall identify candidate groups using a community-detection algorithm (initially Louvain or another feasible alternative).	Must	T
FR-19	The system shall record graph construction and threshold parameters for each run.	Should	I

Table 9. Scoring and explanation

ID	Requirement	Priority	Verif.
FR-20	The system shall compute normalized text, temporal and structural feature scores for each candidate group where the required input fields are available; unavailable signals shall be identified explicitly.	Must	T
FR-21	For groups with sufficient available signals, the system shall combine feature scores into a 0–100 suspicion score using a transparent weighted combination with documented, configurable weights. It shall report which signals were used and shall not treat an incomplete score as equivalent to a full three-signal score.	Must	T
FR-22	The system shall rank candidate groups by their suspicion score.	Must	T
FR-23	For each flagged group, the system shall present the available underlying signals and supporting evidence, such as representative similar comments, overlapping products and temporal activity patterns, where the required data is available.	Must	D
FR-24	The system shall state that the score is a prioritization measure and not proof of fraud or a probability of dishonesty.	Must	I
FR-25	The system could use a learned fusion model with SHAP explanations if reliable labeled training data and sufficient time are available.	Could	A

Table 10. Evaluation

ID	Requirement	Priority	Verif.
FR-26	The system shall run text-only, time-only, graph-only and combined configurations on the same data and splits, supporting an ablation study.	Must	A
FR-27	Where appropriate ground-truth labels permit, the system shall report precision, recall and F1 at the corresponding evaluation level. Precision at top-k flagged groups and ROC-AUC shall be reported only where meaningful labels and definitions permit. Runtime shall also be reported; unavailable metrics shall be identified with reasons.	Must	A
FR-28	If suitable Turkish group-level ground-truth labels are insufficient and an authorized Turkish background dataset is available, the evaluation shall use controlled injection of coordinated groups (identical comments, paraphrased comments and less synchronized posting) and record injected membership. This is an evaluation fallback, not a claim of real-world fraud labels.	Conditional	T
FR-29	If controlled groups are injected, the evaluation should measure group recovery by comparing detected groups with known injected membership.	Should	A
FR-30	Results on real and synthetic data shall be reported separately.	Must	I
FR-31	The system shall collect false-positive examples for error analysis.	Should	A

Table 11. Results API and web interface

ID	Requirement	Priority	Verif.
FR-32	The backend shall provide an API to upload a dataset, start an analysis and retrieve its results.	Must	T
FR-33	The web dashboard shall list the flagged groups ordered by suspicion score.	Must	D
FR-34	The web dashboard shall show a detail view of a selected group with the evidence described in FR-23.	Must	D
FR-35	The web dashboard shall provide a graph view of the selected group and its reviewed products.	Must	D
FR-36	The web dashboard shall provide basic visualizations such as review activity over time.	Should	D

3.3 Non-functional requirements

Table 12. Non-functional requirements

ID	Category	Requirement	Verif.
NFR-01	Explainability	Every suspicion score shall be traceable to its component signals and to the evidence displayed for the group.	I
NFR-02	Reproducibility	Running the same input with the same configuration shall produce the same results, except for documented nondeterministic steps; model versions, parameters, random seeds and software environment shall be recorded. Experiments shall be kept in reproducible notebooks.	T
NFR-03	Privacy	The system shall not require personally identifying data and shall operate on anonymized account identifiers; reports, the webpage and presentations shall contain only aggregated or anonymized examples.	I
NFR-04	Legal compliance	Only authorized datasets with documented provenance and licenses shall be used; live scraping shall not be performed.	I
NFR-05	Human oversight	The system shall not perform automated moderation or enforcement; all outputs are intended for human review.	I
NFR-06	Maintainability	Import, preprocessing, text, temporal, graph, scoring, API and interface components shall be separable and independently testable; the embedding model and community-detection algorithm shall be replaceable.	I

ID	Category	Requirement	Verif.
NFR-07	Performance and scalability	Graph construction shall restrict candidate edges so that analysis remains tractable on the evaluation datasets; runtime shall be measured and reported.	A
NFR-08	Usability	An analyst shall be able to upload data, run an analysis and inspect flagged groups through the web interface.	D
NFR-09	Documentation	Source code, setup instructions and the evaluation dataset protocol shall be documented; reports and backlog versions shall be available on the project webpage.	I

3.4 Data requirements

The input schema includes the following fields. Additional metadata is optional. If an input field is missing, the system will perform only those analyses supported by the available data, report the limitations, and will not infer missing values (FR-04). If the available fields are insufficient for meaningful analysis, the system will report that limitation.

Table 13. Input schema

Field	Description	Analyses limited if missing
review_id	Unique identifier of the review	Identification of individual reviews in results
account_id	Anonymized identifier of the reviewer account	Reviewer–product graph, candidate groups and account-level coordination signals
product_id	Identifier of the reviewed product	Per-product similarity, burst detection and reviewer–product graph
review_text	Text of the review (Turkish)	Text-similarity analysis
rating	Rating given by the reviewer	Rating information in the review record
timestamp	Date and time at which the review was posted	Temporal analysis and timing-based edge weights
other metadata	Optional additional fields provided by the dataset	None

3.5 Implementation constraints (pseudo requirements)

The following implementation choices are provisional; they will be confirmed according to the available datasets, team familiarity and prototype performance.

Table 14. Planned technologies

Component	Planned implementation
Data processing	Python, pandas, NumPy
Text analysis	Sentence Transformers / Hugging Face
Temporal features	pandas and SciPy or simple statistical rules
Graph analysis	NetworkX and one community-detection algorithm
Backend	FastAPI
Frontend	React; optional graph visualization library
Storage	SQLite initially; PostgreSQL if needed
Evaluation	scikit-learn and reproducible notebooks

- The system shall operate in batch mode on uploaded datasets; real-time platform integration is excluded.
- Graph neural networks, web scraping and automated moderation are excluded from the prototype.

4. References

ACM. (2018). *ACM code of ethics and professional conduct.* Association for Computing Machinery. https://www.acm.org/code-of-ethics

ACM/IEEE-CS Joint Task Force on Software Engineering Ethics and Professional Practices. (1999). *Software engineering code of ethics and professional practice* (Version 5.2). https://www.acm.org/code-of-ethics/software-engineering-code

Akkol, E., & Gökşen, Y. (2024). Creating a comprehensive data set for deception detection studies in Turkish texts. *Research Journal of Business and Management, 11*(2), 138–145. https://doi.org/10.17261/Pressacademia.2024.1960

Akoglu, L., Chandy, R., & Faloutsos, C. (2013). Opinion fraud detection in online reviews by network effects. In *Proceedings of the Seventh International AAAI Conference on Weblogs and Social Media (ICWSM 2013).* AAAI Press.

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment, 2008*(10), P10008. https://doi.org/10.1088/1742-5468/2008/10/P10008

Directive (EU) 2019/2161 of the European Parliament and of the Council of 27 November 2019 amending Council Directive 93/13/EEC and Directives 98/6/EC, 2005/29/EC and 2011/83/EU as regards the better enforcement and modernisation of Union consumer protection rules. (2019). *Official Journal of the European Union, L 328*, 7–28. http://data.europa.eu/eli/dir/2019/2161/oj

Dou, Y., Liu, Z., Sun, L., Deng, Y., Peng, H., & Yu, P. S. (2020). Enhancing graph neural network-based fraud detectors against camouflaged fraudsters. In *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM '20)* (pp. 315–324). ACM. https://doi.org/10.1145/3340531.3411903

Erdem & Erdem. (2026). *Ticari Reklam ve Haksız Ticari Uygulamalar Yönetmeliği'nde kapsamlı değişiklikler yapıldı.* https://www.erdem-erdem.av.tr/bilgi-bankasi/ticari-reklam-ve-haksiz-ticari-uygulamalar-yonetmeliginde-kapsamli-degisiklikler-yapildi

Fei, G., Mukherjee, A., Liu, B., Hsu, M., Castellanos, M., & Ghosh, R. (2013). Exploiting burstiness in reviews for review spammer detection. In *Proceedings of the Seventh International AAAI Conference on Weblogs and Social Media (ICWSM 2013).* AAAI Press.

Hooi, B., Song, H. A., Beutel, A., Shah, N., Shin, K., & Faloutsos, C. (2016). FRAUDAR: Bounding graph fraud in the face of camouflage. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 895–904). ACM. https://doi.org/10.1145/2939672.2939747

IEEE. (n.d.). *IEEE code of ethics.* Institute of Electrical and Electronics Engineers. https://www.ieee.org/about/corporate/governance/p7-8.html

Jindal, N., & Liu, B. (2008). Opinion spam and analysis. In *Proceedings of the 2008 International Conference on Web Search and Data Mining (WSDM '08)* (pp. 219–230). ACM. https://doi.org/10.1145/1341531.1341560

Kişisel Verilerin Korunması Kanunu [Law on the Protection of Personal Data], Law No. 6698. (2016). *Resmî Gazete* [Official Gazette], No. 29677, 7 April 2016.

Mukherjee, A., Liu, B., & Glance, N. (2012). Spotting fake reviewer groups in consumer reviews. In *Proceedings of the 21st International Conference on World Wide Web* (pp. 191–200). ACM.

Mukherjee, A., Venkataraman, V., Liu, B., & Glance, N. (2013). What Yelp fake review filter might be doing? In *Proceedings of the Seventh International AAAI Conference on Weblogs and Social Media (ICWSM 2013).* AAAI Press.

Ott, M., Choi, Y., Cardie, C., & Hancock, J. T. (2011). Finding deceptive opinion spam by any stretch of the imagination. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies* (pp. 309–319). Association for Computational Linguistics. https://aclanthology.org/P11-1032

Para Dergi. (2026, October 14). *E-ticarette sahte beğeni ekonomisine darbe.* https://www.paradergi.com.tr/teknoloji/2026/10/14/e-ticarette-sahte-begeni-ekonomisine-darbe

Rayana, S., & Akoglu, L. (2015). Collective opinion spam detection: Bridging review networks and metadata. In *Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 985–994). ACM. https://doi.org/10.1145/2783258.2783370

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)* (pp. 3982–3992). Association for Computational Linguistics. https://doi.org/10.18653/v1/D19-1410

Reimers, N., & Gurevych, I. (2020). Making monolingual sentence embeddings multilingual using knowledge distillation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (pp. 4512–4525). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.emnlp-main.365

Tüketicinin Korunması Hakkında Kanun [Law on Consumer Protection], Law No. 6502. (2013). *Resmî Gazete* [Official Gazette], No. 28835, 28 November 2013.

Wang, G., Xie, S., Liu, B., & Yu, P. S. (2011). Review graph based online store review spammer detection. In *Proceedings of the 2011 IEEE 11th International Conference on Data Mining* (pp. 1242–1247). IEEE. https://doi.org/10.1109/ICDM.2011.124

Xie, S., Wang, G., Lin, S., & Yu, P. S. (2012). Review spam detection via temporal pattern discovery. In *Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 823–831). ACM. https://doi.org/10.1145/2339530.2339662

Yu, S., Ren, J., Li, S., Naseriparsa, M., & Xia, F. (2022). Graph learning for fake review detection. *Frontiers in Artificial Intelligence, 5*, Article 922589. https://doi.org/10.3389/frai.2022.922589